Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/05/2026
PI Name:	Landon, Lauren Blackwell Ph.D.		
Project Title:	Communication Delays in Cislunar Space: A Lab Study Examining Team Risk Concerns		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (3) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2024	**End Date:**	08/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2025 per L. Dang/HFBP (Ed., 6/10/25).		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI), Jessica Marquez, Ph.D. has been added as a Co-Investigator. Dr. Marquez is based at NASA Ames Research Center.		
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center) Marquez, Jessica (NASA Ames)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts on lunar missions will maintain communications with ground support through a communications network that is expected to have communication (comm) delays. While at this time, the exact range of these delays are currently unknown, they seem to be in the range of 2 seconds to 12 seconds one-way, but may vary depending on the criticality of the type of communication at hand. Unfortunately, there is a lack of data testing the effects of 2-12 second one-way comm delays. The Apollo missions space-to-ground audio comm delay was <1.5s one-way, but future missions will use different systems, and current NASA experts project longer delays as we move beyond low Earth orbit. While recent analog studies through the NASA Human Research Program (HRP) have evaluated comm delay effects on performance, much of this work has been completed in context of a Mars mission with comm delay on the order of minutes, not seconds. In these past comm delay studies, findings have shown detrimental effects on individual well-being and performance, team and multi-team cohesion and performance, and family connectedness. Effects have been found across a range of tasks (i.e., nominal, maintenance, extravehicular activities (EVAs), troubleshooting, contingency, and medical/emergency), but again, not investigated at a lunar comm delay. NASA experts have also stated that current countermeasures will not be effective in beyond low Earth orbit (BLEO); for example, the VSee videoconferencing software used for private medical conferences does not work with current projected comm delay latencies, and remote guidance can be significantly more difficult. The current engineering paradigm relies on ground expertise and real-time communication for most analysis and repair activities, which will not be possible with lunar operations. Later Artemis missions will involve many planned team tasks that are dangerous and complex, such as EVAs, which are currently closely monitored and coordinated from Earth. Longer duration lunar missions may also see cumulative effects of comm delay frustrations; and comm quality declines disrupt shared mental models and space-to-ground cohesion. However, we do not understand where “breaking points” exist for different space-to-ground team operations and when well-being and team cohesion will be significantly decremented. It is critical we examine the impact of lunar comm delays on team performance and cohesion as the Artemis missions take us back to the Moon.
Rationale for HRP Directed Research:	Results from this assessment will be used to help set the likely range of lunar comm delay studied in the NASA Human Exploration Research Analog (HERA) Campaign 8. Given that HERA Campaign 8 planning will need to be defined in about 15 months, there is not enough time to solicit and complete this pre-cursor effort. As a result, HFBP is initiating this task as a directed task.
Research Impact/Earth Benefits:	Astronauts on lunar missions will maintain communications with ground support through a communications network that is expected to have communication (comm) delays. While at this time, the exact range of these delays are currently unknown, they seem to be in the realm of 4 seconds to 12 seconds one way, but may vary depending on the criticality of the type of communication at hand. While recent analog studies through HRP have evaluated comm delay effects on performance, much of this work has been completed in context of a Mars mission – meaning extended delays of +/- five minutes one-way. As NASA focuses on upcoming lunar missions, comm delay is still expected to play a part, but there is a lack of data under comm delays of 2-14 seconds. While the effects of this degree of comm delay on space-to-ground coordination are less understood, the implications may be larger than realized, particularly for time-pressured or off nominal situations. The purpose of this effort is to examine the likely range of lunar comm delays and how they may affect team-related performance outcomes, identify where ‘breaking points’ may occur, and understand how teams may respond to those delays. Results from this assessment will be used to help set the likely range of lunar comm delay studied in HERA Campaign 8.
Task Progress:	The research team reviewed and discussed the team articles collected in a related HFBP task (State of Communication Delay Research). Note: this study is closely coordinating with the directed task led by PI Tina Holden, Ph.D., “Communication Delays in Cislunar Space: A Lab Study Examining Team Risk Concerns.” The research team is a collective of both projects’ teams working together. During last year, the research team collected data with the flight-like prototype interface to examine performance and problem solving of a crewmember/flight controller team collaborating to perform realistic spaceflight tasks under different levels of lunar communication delays. The prototypes provided automated data collection capability (i.e., click recording and timing). Qualtrics surveys were used to collect demographic and background information, as well as to collect self-reported ratings of workload, team processes, shared mental models, comm quality, delay frustration, individual and team performance, autonomy and reliance on MCC, and cohesion, as well as repeated baseline measures of mood, stress, and fatigue. Objective measures of team task performance included number of errors, time to complete, and percent of objective completed. Communication quantity metrics include number of words and messages sent. Data collection ended in June 2025. The 46 participants (after 2 withdraws) completed each session with a partner. Partners could be new or repeat partners. Some participants did not experience all 6 conditions. Each session was conducted under one of the delay conditions and included the participant pair performing one task under nominal conditions and another task under off-nominal conditions. Results indicated that communication delays resulted in decreased scores for measures of team processes, cohesion, and performance, and increased stress and frustration, when compared to real time communications. Longer delays, often starting at 6s or greater, were worse than no delays or short 4s delays. Off-nominal tasks saw worse team outcomes than nominal tasks, while participants in the MCC role (who had stale data and was reliant on crew information) had lower ratings than participants in the crew role. There were several 2- and 3-way interactions with comm delays, task type, and participant role. Results varied by outcome. Details are included in the final report to the funding Element.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Holden K, Landon LB, Upasani S, Moody N, Duke T, Mosonyi J, Jorge M, Marquez,JJ, Karasinski J. "The impact of lunar communication delays on individual and team performance." Human Factors and Ergonomics Society, Southern Regional Conference Meeting, Houston, TX, April 11, 2025. Abstracts. Human Factors and Ergonomics Society, Southern Regional Conference Meeting, Houston, TX, April 11, 2025. , Apr-2025

Abstracts for Journals and Proceedings	Upasani S, Mosonyi J, Moody N, Duke T, Jorge M, Marquez JJ, Karasinski J, Landon LB, Holden K. "Communication delays in cislunar space: A lab study examining HSIA/Team risk concerns." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025